Extended Kalman Filter:

Application to IM with Ttype MPC Three Phase Inverter.

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In this document the model of a generic flexible shaft is derived.

1 Induction Motor Model Derivation

In this section we derive the dynamical model of the induction motor for control purpose. For the derivation of the dynamical model, the rotor windings are assumed to be identical to the stator windings with resistance R_r (R_s for the stator winding) and equivalent turns $N_r = N_s$.

Let

$$\vec{\psi}^s_{abc} = [\psi_{sa},\, \psi_{sb},\, \psi_{sc}]^T \quad \text{stator flux} \tag{1.1}$$

$$\vec{\psi}^r_{abc} = [\psi_{ra},\, \psi_{rb},\, \psi_{rc}]^T \quad \text{rotor flux} \tag{1.2}$$

be the vectors whose components are the stator and rotor flux linkages, respectively, with a, b, c denoting the three phases components. Similarly, let

$$\begin{aligned} \vec{i}^s_{abc} &= [i_{sa},\, i_{sb},\, i_{sc}]^T \quad \text{stator current} \\ \vec{i}^r_{abc} &= [i_{ra},\, i_{rb},\, i_{rc}]^T \quad \text{rotor current} \end{aligned} \tag{1.3}$$

be the vectors whose components are the stator and rotor currents. Then, for an induction motor with one pole pair we can write

$$\left.\begin{aligned} \frac{d\psi_{sa}}{dt} + R_s i_{sa} &= u_{sa} \\ \frac{d\psi_{sb}}{dt} + R_s i_{sb} &= u_{sb} \\ \frac{d\psi_{sc}}{dt} + R_s i_{sc} &= u_{sc} \end{aligned}\right\} \quad \text{respect to the stationary reference frame}$$

$$\left.\begin{aligned} \frac{d\psi_{ra}}{dt} + R_r i_{ra} &= 0 \\ \frac{d\psi_{rb}}{dt} + R_r i_{rb} &= 0 \\ \frac{d\psi_{rc}}{dt} + R_r i_{rc} &= 0 \end{aligned}\right\} \quad \text{respect to the rotor reference frame} \tag{1.4}$$

where the stator and rotor fluxes, under the assumption of linear magnetic circuits, satisfy the linear relation

$$\begin{bmatrix} \vec{\psi}^s_{abc} \\ \vec{\psi}^r_{abc} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_m \\ \mathbf{L}_m^T & \mathbf{L}_r \end{bmatrix} \begin{bmatrix} \vec{i}^s_{abc} \\ \vec{i}^r_{abc} \end{bmatrix} \tag{1.5}$$

where

$$\mathbf{L}_s = \begin{bmatrix} L_\mu + L_s & -\frac{1}{2}L_s & -\frac{1}{2}L_s \\ -\frac{1}{2}L_s & L_\mu + L_s & -\frac{1}{2}L_s \\ -\frac{1}{2}L_s & -\frac{1}{2}L_s & L_\mu + L_s \end{bmatrix} \tag{1.6}$$

$$\mathbf{L}_r = \begin{bmatrix} L_\mu + L_r & -\frac{1}{2}L_r & -\frac{1}{2}L_r \\ -\frac{1}{2}L_r & L_\mu + L_r & -\frac{1}{2}L_r \\ -\frac{1}{2}L_r & -\frac{1}{2}L_r & L_\mu + L_r \end{bmatrix} \tag{1.7}$$

and

$$\mathbf{L}_m = L_m \begin{bmatrix} \cos(\vartheta_m) & \cos(\vartheta_m + \frac{2\pi}{3}) & \cos(\vartheta_m - \frac{2\pi}{3}) \\ \cos(\vartheta_m - \frac{2\pi}{3}) & \cos(\vartheta_m) & \cos(\vartheta_m + \frac{2\pi}{3}) \\ \cos(\vartheta_m + \frac{2\pi}{3}) & \cos(\vartheta_m - \frac{2\pi}{3}) & \cos(\vartheta_m) \end{bmatrix} \tag{1.8}$$

Considering the transformation

$$\mathbf{T}_1 = \frac{2}{3} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \tag{1.9}$$

such that

$$\begin{bmatrix} i_{s0} \\ i_{s\alpha} \\ i_{s\beta} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_{sa} \\ i_{sb} \\ i_{sc} \end{bmatrix} \tag{1.10}$$

in which $\mathbf{T}_1^{-1} = \mathbf{T}_1^T$.

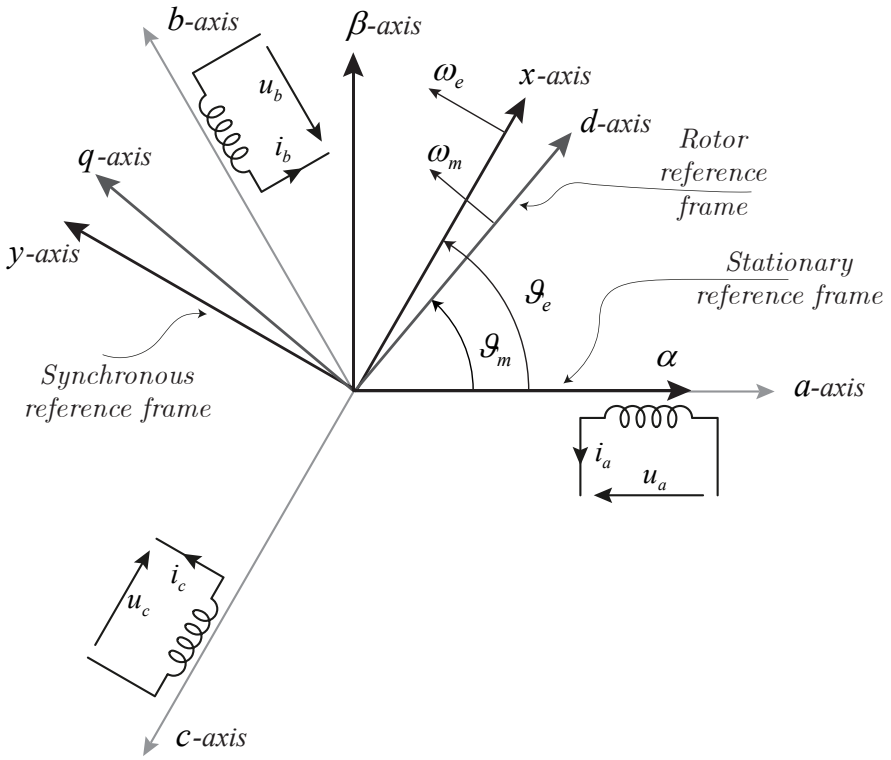


Figure 1: Machine reference frames description: stationary (α, β), rotor (d, q) and synchronous speed (x, y) reference frames.

The Eq. Eq. (1.4) of the induction motor become (considering balanced operating conditions, $i_{s0} = 0$, $i_{r0} = 0$)

$$\left.\begin{aligned} \frac{d\psi_{s\alpha}}{dt} + R_s i_{s\alpha} &= u_{s\alpha} \\ \frac{d\psi_{s\beta}}{dt} + R_s i_{s\beta} &= u_{s\beta} \end{aligned}\right\} \text{ stationary reference frame}$$

$$\left.\begin{aligned} \frac{d\psi_{rd}}{dt} + R_r i_{rd} &= 0 \\ \frac{d\psi_{rq}}{dt} + R_r i_{rq} &= 0 \end{aligned}\right\} \text{ rotor reference frame} \tag{1.11}$$

$$\left.\begin{aligned} \dot{\vartheta} &= \omega_m \\ J\dot{\omega}_m &= \tau_e - \tau_l \end{aligned}\right\} \text{ mechanical equations}$$

where (ψ_{rd}, ψ_{rq}) and (i_{rd}, i_{rq}) denote the (d, q) components of the rotor flux and rotor current in the (d, q) reference frame attached to the rotor, rotating at rotor speed $\omega_m = \dot{\vartheta}_m$, while ($\psi_{s\alpha}$, $\psi_{s\beta}$) and ($u_{s\alpha}$, $u_{s\beta}$) denote the components of the stator flux and the stator voltage vectors in the fixed reference frame.

The linear operator for changing of coordinates between stationary reference frame and rotor reference



frame is

$$\mathbf{T}_2(\vartheta_m) = \begin{bmatrix} \cos\vartheta_m & \sin\vartheta_m \\ -\sin\vartheta_m & \cos\vartheta_m \end{bmatrix} \qquad \mathbf{T}_2^{-1}(\vartheta_m) = \begin{bmatrix} \cos\vartheta_m & -\sin\vartheta_m \\ \sin\vartheta_m & \cos\vartheta_m \end{bmatrix} \tag{1.12}$$

such that

$$\begin{bmatrix} i_{rd} \\ i_{rq} \end{bmatrix} = \begin{bmatrix} \cos\vartheta_m & \sin\vartheta_m \\ -\sin\vartheta_m & \cos\vartheta_m \end{bmatrix} \begin{bmatrix} i_{r\alpha} \\ i_{r\beta} \end{bmatrix} \tag{1.13}$$

$$\begin{bmatrix} \psi_{rd} \\ \psi_{rq} \end{bmatrix} = \begin{bmatrix} \cos\vartheta_m & \sin\vartheta_m \\ -\sin\vartheta_m & \cos\vartheta_m \end{bmatrix} \begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} \tag{1.14}$$

with $(\psi_{r\alpha}, \psi_{r\beta})$ and $(i_{r\alpha}, i_{r\beta})$ denoting the (α, β)-components of the rotor flux and current in the fixed frame.

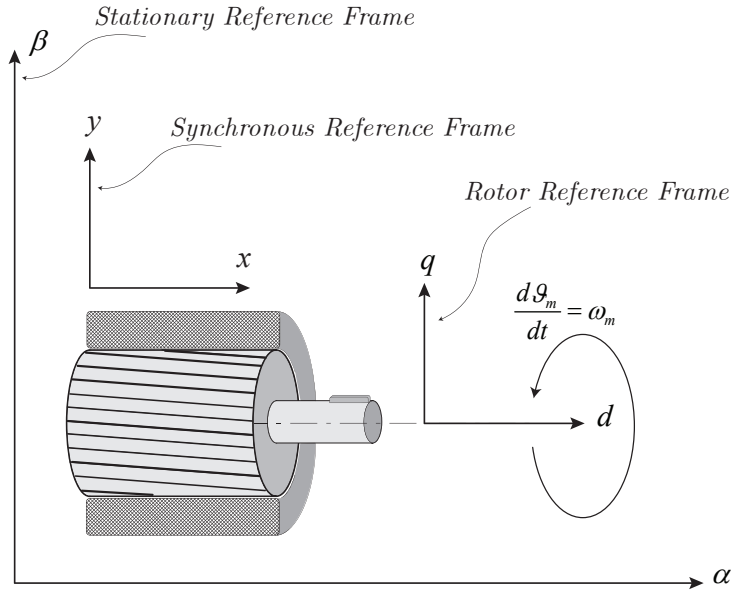


Figure 2: Different point of views regarding different reference frames: stationary (α, β) can be considered in still condition; rotor (d, q) can be thought rotating at the rotor speed; synchronous reference frame (x, y) can be thought rotating at electrical stator pulsation.

As we have seen, stator flux equations are represented respect the stationary reference frame while the rotor flux equations are represented respect to the rotor reference frame. Now we want to transform the rotor flux equations from the rotor to the stationary reference frame.

Considering that

$$\frac{d\mathbf{T}_2(\vartheta_m)}{dt} = -\omega_m \mathbf{T}_2(\vartheta_m) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\omega_m \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{T}_2(\vartheta_m) \tag{1.15}$$

$$\frac{d\mathbf{T}_2^{-1}(\vartheta_m)}{dt} = -\omega_m \mathbf{T}_2^{-1}(\vartheta_m) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\omega_m \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{T}_2^{-1}(\vartheta_m) \tag{1.16}$$

The rotor flux equations respect to the stationary reference frame become as follows

$$\begin{aligned} \dot{\vec{\psi}}_{dq}^{r} = \frac{d}{dt}\left(\mathbf{T}_2(\vartheta_m)\vec{\psi}_{\alpha\beta}^{r}\right) &= -\omega_m \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{T}_2(\vartheta_m)\vec{\psi}_{\alpha\beta}^{r} + \mathbf{T}_2(\vartheta_m)\dot{\vec{\psi}}_{\alpha\beta}^{r} \\ &= -\mathbf{R}_r \mathbf{T}_2(\vartheta_m)\vec{i}_{\alpha\beta}^{r} \end{aligned} \tag{1.17}$$

multiplying left side by $\mathbf{T}_2^{-1}(\vartheta_m)$ we obtain

$$\dot{\vec{\psi}}_{\alpha\beta}^{r} = \omega_m \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \vec{\psi}_{\alpha\beta}^{r} - \mathbf{R}_r \vec{i}_{\alpha\beta}^{r} \tag{1.18}$$

or

$$\begin{bmatrix} \dot{\psi}_{r\alpha} \\ \dot{\psi}_{r\beta} \end{bmatrix} = - \begin{bmatrix} R_r & 0 \\ 0 & R_r \end{bmatrix} \begin{bmatrix} i_{r\alpha} \\ i_{r\beta} \end{bmatrix} - \begin{bmatrix} 0 & \omega_m \\ -\omega_m & 0 \end{bmatrix} \begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} \tag{1.19}$$

Now we can write all terms of the Eq. Eq. (1.11) in (α, β) coordinates as follows

$$\begin{aligned} \begin{bmatrix} \dot{\psi}_{s\alpha} \\ \dot{\psi}_{s\beta} \end{bmatrix} &= - \begin{bmatrix} R_s & 0 \\ 0 & R_s \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} + \begin{bmatrix} u_{s\alpha} \\ u_{s\beta} \end{bmatrix} \\ \begin{bmatrix} \dot{\psi}_{r\alpha} \\ \dot{\psi}_{r\beta} \end{bmatrix} &= - \begin{bmatrix} R_r & 0 \\ 0 & R_r \end{bmatrix} \begin{bmatrix} i_{r\alpha} \\ i_{r\beta} \end{bmatrix} - \begin{bmatrix} 0 & \omega_m \\ -\omega_m & 0 \end{bmatrix} \begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} \end{aligned} \tag{1.20}$$

The electromagnetic equations are

$$\begin{bmatrix} \psi_{s\alpha} \\ \psi_{s\beta} \\ \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} = \begin{bmatrix} L_s & 0 & L_m & 0 \\ 0 & L_s & 0 & L_m \\ L_m & 0 & L_r & 0 \\ 0 & L_m & 0 & L_r \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \\ i_{r\alpha} \\ i_{r\beta} \end{bmatrix} = \mathbf{L} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \\ i_{r\alpha} \\ i_{r\beta} \end{bmatrix} \tag{1.21}$$

i.e. the quadratic form associated to $\mathbf{L}$

$$\frac{3}{2}\left(\frac{1}{2}\vec{i}_{\alpha\beta}^{\,T}\mathbf{L}\vec{i}_{\alpha\beta}\right) \tag{1.22}$$

is positive for any non-zero value of the current vector $\vec{i}_{\alpha\beta} = [i_{s\alpha}, i_{s\beta}, i_{r\alpha}, i_{r\beta}]$ and represent the magnetic energy.

Define

$$\mathbf{R} = \begin{bmatrix} R_s & 0 & 0 & 0 \\ 0 & R_s & 0 & 0 \\ 0 & 0 & R_r & 0 \\ 0 & 0 & 0 & R_r \end{bmatrix} \tag{1.23}$$

the state space model representation of the system Eq. Eq. (1.20) is

$$\begin{aligned} \mathbf{L} \begin{bmatrix} \frac{d}{dt} i_{s\alpha} \\ \frac{d}{dt} i_{s\beta} \\ \frac{d}{dt} i_{r\alpha} \\ \frac{d}{dt} i_{r\beta} \end{bmatrix} &= -\mathbf{R} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \\ i_{r\alpha} \\ i_{r\beta} \end{bmatrix} + \\ &+ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \omega_m L_m & 0 & \omega_m L_r \\ -\omega_m L_m & 0 & -\omega_m L_r & 0 \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \\ i_{r\alpha} \\ i_{r\beta} \end{bmatrix} + \begin{bmatrix} u_{s\alpha} \\ u_{s\beta} \\ 0 \\ 0 \end{bmatrix} \\ J \frac{d}{dt} \omega_m &= \tau_e - \tau_l \end{aligned} \tag{1.24}$$

in which the state variables are ($i_{s\alpha}$, $i_{s\beta}$, $i_{r\alpha}$, $i_{r\beta}$, ω_m) and the electromagnetic torque τ_e is still to be determined as a function of the state variables. This choice of state variables is naturally linked to the energy stored in the energy stored in the motor given by

$$E = \frac{3}{2} \left(\frac{1}{2} \vec{i}_{\alpha\beta}^{\,T} \mathbf{L} \, \vec{i}_{\alpha\beta} \right) + \frac{1}{2} J \omega_m^2 \tag{1.25}$$

which is the sum of the magnetic energy and of the kinetic energy

$$\frac{1}{2} J \omega_m^2 \tag{1.26}$$

The expression of τ_e can be obtained from the energy balance

$$\begin{aligned} \frac{dE}{dt} =& \frac{3}{2} \begin{bmatrix} i_{s\alpha} & i_{s\beta} \end{bmatrix} \begin{bmatrix} u_{s\alpha} & u_{s\beta} \end{bmatrix}^T - \tau_l \, \omega_m - \frac{3}{2} \vec{i}_{\alpha\beta}^{\,T} \mathbf{R} \vec{i}_{\alpha\beta} + \\ &+ \frac{3}{2} \omega_m \begin{bmatrix} i_{r\alpha} & i_{r\beta} \end{bmatrix} \begin{bmatrix} 0 & -L_m \\ L_m & 0 \end{bmatrix} \begin{bmatrix} i_{s\alpha} & i_{s\beta} \end{bmatrix}^T + \tau_e \omega_m \end{aligned} \tag{1.27}$$

Let P_{in}, P_{out} and P_{loss} denote the input power, the output power, and the power losses, respectively;

since

$$\begin{aligned}\frac{dE}{dt} =& P_{in} - P_{out} - P_{loss} \\ =& \frac{3}{2}\begin{bmatrix} i_{s\alpha} & i_{s\beta} \end{bmatrix}\begin{bmatrix} u_{s\alpha} & u_{s\beta} \end{bmatrix}^T - \tau_l\,\omega_m - \frac{3}{2}\vec{i}_{\alpha\beta}^{\,T}\mathbf{R}\vec{i}_{\alpha\beta}\end{aligned} \tag{1.28}$$

which means

$$\tau_e = \frac{3}{2}\begin{bmatrix} i_{r\alpha} & i_{r\beta} \end{bmatrix}\begin{bmatrix} 0 & -L_m \\ L_m & 0 \end{bmatrix}\begin{bmatrix} i_{s\alpha} & i_{s\beta} \end{bmatrix}^T = \frac{3}{2}L_m\left(i_{r\alpha}i_{s\beta} - i_{r\beta}i_{s\alpha}\right) \tag{1.29}$$

Note that the electromagnetic torque τ_e produced by the motor is a non-linear function of the state variables and constitutes the main non-linear term in the induction motor model. Since

$$\begin{aligned} i_{r\alpha} &= -\frac{L_m}{L_r}i_{s\alpha} + \frac{1}{Lr}\psi_{r\alpha} \\ i_{r\beta} &= -\frac{L_m}{L_r}i_{s\beta} + \frac{1}{Lr}\psi_{r\beta}\end{aligned} \tag{1.30}$$

the electromagnetic torque can be expressed as

$$\tau_e = \frac{3}{2}\frac{L_m}{L_r}\left(\psi_{r\alpha}i_{s\beta} - \psi_{r\beta}i_{s\alpha}\right). \tag{1.31}$$

Eliminating $(\psi_{s\alpha}, \psi_{s\beta}, i_{r\alpha}, i_{r\beta})$ from equation 1.20 using equations 1.20 and 1.30, namely

$$\begin{bmatrix} i_{r\alpha} \\ i_{r\beta} \end{bmatrix} = \begin{bmatrix} -\frac{L_m}{L_r} & 0 \\ 0 & -\frac{L_m}{L_r} \end{bmatrix}\begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_r} & 0 \\ 0 & \frac{1}{L_r} \end{bmatrix}\begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} \tag{1.32}$$

$$\begin{bmatrix} \psi_{s\alpha} \\ \psi_{s\beta} \end{bmatrix} = \begin{bmatrix} L_s - \frac{L_m^2}{L_r} & 0 \\ 0 & L_s - \frac{L_m^2}{L_r} \end{bmatrix}\begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} + \begin{bmatrix} \frac{L_m}{L_r} & 0 \\ 0 & \frac{L_m}{L_r} \end{bmatrix}\begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix} \tag{1.33}$$

we obtain a state model in terms of the state variables $(\omega_m,\, \psi_{r\alpha},\, \psi_{r\beta},\, i_{s\alpha},\, i_{s\beta})^T$, the input variables $(u_{s\alpha},\, u_{s\beta})^T$ and the disturbance τ_l

IM Dynamical Model Respect to a Stationary Reference Frame

$$\begin{aligned}
\frac{di_{s\alpha}}{dt} &= -\gamma i_{s\alpha} + \beta\alpha\psi_{r\alpha} + \beta\omega_m\psi_{r\beta} + \frac{u_{s\alpha}}{\sigma} \\
\frac{di_{s\beta}}{dt} &= -\gamma i_{s\beta} + \beta\alpha\psi_{r\beta} - \beta\omega_m\psi_{r\alpha} + \frac{u_{s\beta}}{\sigma} \\
\frac{d\psi_{r\alpha}}{dt} &= -\alpha\psi_{r\alpha} - \omega_m\psi_{r\beta} + \alpha L_m i_{s\alpha} \\
\frac{d\psi_{r\beta}}{dt} &= -\alpha\psi_{r\beta} + \omega_m\psi_{r\alpha} + \alpha L_m i_{s\beta} \\
\frac{d\omega_m}{dt} &= \mu\left(\psi_{r\alpha} i_{s\beta} - \psi_{r\beta} i_{s\alpha}\right) - \frac{\tau_l}{J}
\end{aligned} \tag{1.34}$$

where

$$\begin{aligned}
&\mu = \frac{3}{2}\frac{L_m}{JL_r}, \quad \alpha = \frac{R_r}{L_r}, \quad \sigma = L_s\left(1 - \frac{L_m^2}{L_s L_r}\right) \\
&\beta = \frac{L_m}{\sigma L_r}, \quad \gamma = \frac{R_s}{\sigma} + \beta\alpha L_m
\end{aligned} \tag{1.35}$$

Consider now the condition of constant speed

$$\frac{d\omega_m}{dt} = 0 \quad \Rightarrow \quad \omega_m = \omega_m^0 \tag{1.36}$$

which brings the above system from a nonlinear time-variant into a linear time-invariant as shown in the following

IM Dynamical Model Respect to a Stationary Reference Frame and Constant Rotor Speed

$$\begin{aligned}
\dot{\vec{x}} &= \tilde{\mathbf{A}}\vec{x} + \tilde{\mathbf{B}}\vec{u} \\
\vec{y} &= \mathbf{C}\vec{x}
\end{aligned} \tag{1.37}$$

where

$$\vec{x} = \begin{bmatrix} i_{s\alpha} & i_{s\beta} & \psi_{r\alpha} & \psi_{r\beta} \end{bmatrix}^T$$

$$\vec{u} = \begin{bmatrix} u_{s\alpha} & u_{s\beta} \end{bmatrix}^T$$

$$\vec{y} = \begin{bmatrix} i_{s\alpha} & i_{s\beta} \end{bmatrix}^T$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} -\gamma & 0 & \beta\alpha & \beta\omega_m^0 \\ 0 & -\gamma & -\beta\omega_m^0 & \beta\alpha \\ \alpha L_m & 0 & -\alpha & -\omega_m^0 \\ 0 & \alpha L_m & \omega_m^0 & -\alpha \end{bmatrix}, \qquad \tilde{\mathbf{B}} = \begin{bmatrix} \frac{1}{\sigma} & 0 \\ 0 & \frac{1}{\sigma} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{1.38}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

The expression of the electromagnetic torque is as follows

$$\tau_e = \frac{3}{2}\frac{L_m}{L_r}\Big(\psi_{r\alpha} i_{s\beta} - \psi_{r\beta} i_{s\alpha}\Big) \tag{1.39}$$

The state space model 1.34, which will be referred to as fixed frame model, has some advantages from the control view point since it clarifies that the control inputs ($u_{s\alpha}$, $u_{s\beta}$) directly affect the dynamics of the stator currents ($i_{s\alpha}$, $i_{s\beta}$) which can be viewed as immediate control variables since they control the rotor speed ω_m and the rotor flux modulus $\sqrt{\psi_{r\alpha}^2 + \psi_{r\beta}^2}$ in the first three equations.

Let us now introduce a time-varing (d, q) frame which rotates at an arbitrary speed $\omega_0(t)$ and is identified at each time t by the angle $\vartheta_0(t)$ so that

$$\frac{d\vartheta_0}{dt} = \omega_0 \tag{1.40}$$

Rotor fluxes ($\psi_{r\alpha}$, $\psi_{r\beta}$), stator currents ($i_{s\alpha}$, $i_{s\beta}$), and stator voltages ($u_{s\alpha}$, $u_{s\beta}$) are expressed with respect to the time-varying rotating (d, q) frame as

$$\begin{bmatrix} \psi_{rd} \\ \psi_{rq} \end{bmatrix} = \begin{bmatrix} \cos\vartheta_0 & \sin\vartheta_0 \\ -\sin\vartheta_0 & \cos\vartheta_0 \end{bmatrix} \begin{bmatrix} \psi_{r\alpha} \\ \psi_{r\beta} \end{bmatrix}$$

$$\begin{bmatrix} i_{sd} \\ i_{sq} \end{bmatrix} = \begin{bmatrix} \cos\vartheta_0 & \sin\vartheta_0 \\ -\sin\vartheta_0 & \cos\vartheta_0 \end{bmatrix} \begin{bmatrix} i_{s\alpha} \\ i_{s\beta} \end{bmatrix} \tag{1.41}$$

$$\begin{bmatrix} u_{sd} \\ u_{sq} \end{bmatrix} = \begin{bmatrix} \cos\vartheta_0 & \sin\vartheta_0 \\ -\sin\vartheta_0 & \cos\vartheta_0 \end{bmatrix} \begin{bmatrix} u_{s\alpha} \\ u_{s\beta} \end{bmatrix}$$

If the new **state variables** (ω_m, ψ_{rd}, ψ_{rq}, i_{sd}, i_{sq})^T and input variables (u_{sd}, u_{sq}) are used, in the new



$(d,\, q)$ rotating coordinates the induction motor model becomes; consider the quantity τ_l as disturbance.

IM Dynamical Model Respect to a Rotating Reference Frame

$$\begin{aligned}
\frac{d\omega_m}{dt} &= \mu\left(\psi_{rd} i_{sq} - \psi_{rq} i_{sd}\right) - \frac{\tau_l}{J} \\
\frac{d\psi_{rd}}{dt} &= -\alpha\psi_{rd} + \left(\omega_0 - \omega_m\right)\psi_{rq} + \alpha L_m i_{sd} \\
\frac{d\psi_{rq}}{dt} &= -\alpha\psi_{rq} - \left(\omega_0 - \omega_m\right)\psi_{rd} + \alpha L_m i_{sq} \\
\frac{di_{sd}}{dt} &= -\gamma i_{sd} + \omega_0 i_{sq} + \beta\alpha\psi_{rd} + \beta\omega_m\psi_{rq} + \frac{u_{sd}}{\sigma} \\
\frac{di_{sq}}{dt} &= -\gamma i_{sq} - \omega_0 i_{sd} + \beta\alpha\psi_{rq} - \beta\omega_m\psi_{rd} + \frac{u_{sq}}{\sigma}
\end{aligned} \tag{1.42}$$

where ω_0 is the speed of the rotating reference frame **where**

$$\begin{aligned}
\mu &= \frac{3}{2}\frac{L_m}{JL_r}, \quad \alpha = \frac{R_r}{L_r}, \quad \sigma = L_s\left(1 - \frac{L_m^2}{L_s L_r}\right) \\
\beta &= \frac{L_m}{\sigma L_r}, \quad \gamma = \frac{R_s}{\sigma} + \beta\alpha L_m
\end{aligned} \tag{1.43}$$

In the case the rotating reference is attached to the rotor the induction motor model becomes

IM Dynamical Model Respect to the Rotor Reference Frame

$$\begin{aligned}
\frac{d\omega_m}{dt} &= \mu\left(\psi_{rd} i_{sq} - \psi_{rq} i_{sd}\right) - \frac{\tau_l}{J} \\
\frac{d\psi_{rd}}{dt} &= -\alpha\psi_{rd} + \alpha L_m i_{sd} \\
\frac{d\psi_{rq}}{dt} &= -\alpha\psi_{rq} + \alpha L_m i_{sq} \\
\frac{di_{sd}}{dt} &= -\gamma i_{sd} + \omega_m i_{sq} + \beta\alpha\psi_{rd} + \beta\omega_m\psi_{rq} + \frac{u_{sd}}{\sigma} \\
\frac{di_{sq}}{dt} &= -\gamma i_{sq} - \omega_m i_{sd} + \beta\alpha\psi_{rq} - \beta\omega_m\psi_{rd} + \frac{u_{sq}}{\sigma}
\end{aligned} \tag{1.44}$$

The rotor frame state space model has some advantages from observability view point given by the linearity of the rotor flux equation where the cross terms are deleted from the selected rotating reference frame.



1.0.1 Model discretization

As already mentioned, in order to implement our control structure a discrete-time representation of the system of Eq. Eq. (**??**) shall be derived. By the use of the Euler approximation for the derivatives using the sampling time t_s, that is

$$\frac{d}{dt}x(t) \approx \frac{x[(k+1)t_s] - x(kt_s)}{t_s} = \frac{x(k+1) - x(k)}{t_s} \tag{1.45}$$

we obtain the following discrete time representation of the PMSM model

Discrete-time representation of the PMSM

$$\begin{cases} i_\alpha(k+1) = \Big(1 - t_s\frac{R_s}{L_\alpha}\Big)i_\alpha(k) + t_s\frac{\omega(k)}{L_\alpha}\psi_\beta(k) + \frac{t_s}{L_\alpha}u_\alpha(k) \\ i_\beta(k+1) = \Big(1 - t_s\frac{R_s}{L_\beta}\Big)i_\beta(k) - t_s\frac{\omega(k)}{L_\beta}\psi_\alpha(k) + \frac{t_s}{L_\beta}u_\beta(k) \\ \psi^r_\alpha(k+1) = \psi^r_\alpha(k) - t_s\omega(k)\psi^r_\beta(k) \\ \psi^r_\beta(k+1) = \psi^r_\beta(k) + t_s\omega(k)\psi^r_\alpha(k) \\ \omega_m(k+1) = \omega_m(k) + \frac{t_s}{J}\left[\frac{3}{2}p\Big[\psi^r_\alpha(k)i_\beta(k) - \psi^r_\beta(k)i_\alpha(k)\Big] - \tau_l(k) - b\omega_m(k)\right] \\ \theta(k+1) = \theta(k) + t_s\omega(k) = \theta(k) + t_sp\omega_m(k) \end{cases} \tag{1.46}$$

where p is the number of poles pairs, b the bearing viscosity and $\tau_l(t)$ the external and unknown load (disturbance).

The model of Eq. Eq. (1.46) can be used to implement the MPC as well as the EKF, anyway to implement the EKF we proceed in a little bit different way: we first propose a linearization of Eq. Eq. (**??**) along the estimated state vector $\hat{\vec{x}}$ (where we obtain a linear model) and after that we discretize by Taylor as already seen. This procedure will be described in section 1.1

1.1 Extended Kalman Filter (EKF) Observer

In this section an application of the Extended Kalman filter is presented, where a PMSM rotor fluxes estimator is proposed and implemented. We will proceed using the $\alpha\beta$ model representation reported in Eq. Eq. (**??**).

A different, but equivalent, representation of the PMSM can be re-written as follows

$$\begin{cases} \frac{d}{dt}\vec{x}(t) = \vec{g}(\vec{x}) + \mathbf{B}\vec{u}(t) \\ \vec{y}(t) = \mathbf{C}\vec{x}(t) \end{cases} \tag{1.47}$$



where

$$\vec{x} = \begin{bmatrix} i_\alpha \\ i_\beta \\ \psi_\alpha^r \\ \psi_\beta^r \\ \vartheta \\ \omega \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} \tag{1.48}$$

and where

$$\mathbf{B} = \frac{1}{L_s} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{1.49}$$

and where

$$\frac{d}{dt}\vec{x}(t) = \vec{g}(\vec{x}) \Rightarrow \begin{cases} \frac{d}{dt} i_\alpha(t) = -\frac{R_s}{L_s} i_\alpha(t) + \frac{1}{L_s}\omega(t)\psi_\beta^r(t) + \frac{1}{L_s} u_\alpha(t) \\ \frac{d}{dt} i_\beta(t) = -\frac{R_s}{L_s} i_\beta(t) - \frac{1}{L_s}\omega(t)\psi_\alpha^r(t) + \frac{1}{L_s} u_\beta(t) \\ \frac{d}{dt}\psi_\alpha^r(t) = -\omega(t)\psi_\beta^r(t) \\ \frac{d}{dt}\psi_\beta^r(t) = \omega(t)\psi_\alpha^r(t) \\ \frac{d}{dt}\vartheta(t) = \omega(t) \\ \frac{d}{dt}\omega(t) = \frac{1}{J}\left[\frac{3}{2}\Big(\psi_\alpha^r(t) i_\beta(t) - \psi_\beta^r(t) i_\alpha(t)\Big) - \frac{1}{p}\tau_l(t) - \frac{1}{p} b\omega_m(t)\right] \end{cases} \tag{1.50}$$

we are considering $L_\alpha = \text{Ł}_\beta = L_s$.

For the purpose of the EKF implementation a reduced dynamic description is taken into account, in particular it is assumed the dynamic of the mechanical system is more then one order slower than the electrical system and for this assumption we decide to represent the system of



Eq. Eq. (1.50) as follows

$$\frac{d}{dt}\vec{x}(t) = \vec{f}(\vec{x}) \Rightarrow \begin{cases} \frac{d}{dt}i_\alpha(t) = -\frac{R_s}{L_s}i_\alpha(t) + \frac{1}{L_s}\omega(t)\psi_\beta^r(t) + \frac{1}{L_s}u_\alpha(t) \\ \frac{d}{dt}i_\beta(t) = -\frac{R_s}{L_s}i_\beta(t) - \frac{1}{L_s}\omega(t)\psi_\alpha^r(t) + \frac{1}{L_s}u_\beta(t) \\ \frac{d}{dt}\psi_\alpha^r(t) = -\omega(t)\psi_\beta^r(t) \\ \frac{d}{dt}\psi_\beta^r(t) = \omega(t)\psi_\alpha^r(t) \\ \frac{d}{dt}\vartheta(t) = \omega(t) \\ \frac{d}{dt}\omega(t) = 0 \end{cases} \tag{1.51}$$

where the mechanical dynamic has been **intentionally** approximated to a stationary motion.

In order to implement the EKF we now linearize the system of Eq. Eq. (1.51) along the estimated state space trajectory $\hat{\vec{x}}$, as follows

$$\left.\frac{\partial}{\partial\vec{x}}\vec{f}(\vec{x})\right|_{\vec{x}=\hat{\vec{x}}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{\hat{\vec{x}}(t)} = \tilde{\mathbf{A}}_1 + \hat{\omega}(t)\tilde{\mathbf{A}}_2 + \hat{\psi}_\alpha^r(t)\tilde{\mathbf{A}}_3 + \hat{\psi}_\beta^r(t)\tilde{\mathbf{A}}_4 = \tilde{\mathbf{A}}(t) \tag{1.52}$$

where $\hat{\vec{x}}$ is the estimated state vector from the EKF.

The matrices $\tilde{\mathbf{A}}_1$, $\tilde{\mathbf{A}}_2$, $\tilde{\mathbf{A}}_3$ and $\tilde{\mathbf{A}}_4$ are as follows

$$\tilde{\mathbf{A}}_1 = \begin{bmatrix} -\frac{R_s}{L_s} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{R_s}{L_s} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \tilde{\mathbf{A}}_2 = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{L_s} & 0 & 0 \\ 0 & 0 & -\frac{1}{L_s} & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\tilde{\mathbf{A}}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_s} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \tilde{\mathbf{A}}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{L_s} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{1.53}$$

The discretization of the transition matrix $\tilde{\mathbf{A}}(t)$ can be approximated by Taylor as follows

$$\mathbf{A}(k|k-1) = \mathbf{I} + \tilde{\mathbf{A}}_1 t_s + \hat{\omega}(k|k-1)\tilde{\mathbf{A}}_2 t_s + \hat{\psi}_\alpha^r(k|k-1)\tilde{\mathbf{A}}_3 t_s + \hat{\psi}_\beta^r(k|k-1)\tilde{\mathbf{A}}_4 t_s \tag{1.54}$$

The EKF set of equations can be written as follows

The Extended Kalman Filter Algorithm

$$\hat{\vec{x}}\,(k|k) = \mathbf{A}(k|k-1)\hat{\vec{x}}\,(k|k-1) + \mathbf{B}\vec{u}\,(k|k-1) \tag{1.55}$$

$$\mathbf{P}\,(k|k) = \mathbf{A}(k|k-1)\mathbf{P}\,(k|k-1)\,\mathbf{A}(k|k-1)^T + \mathbf{Q} \tag{1.56}$$

$$\mathbf{L}(k) = \mathbf{P}(k|k)\mathbf{C}^T\Big(\mathbf{C}\,\mathbf{P}(k|k)\,\mathbf{C}^T + \mathbf{R}\Big)^{-1} \tag{1.57}$$

$$\hat{\vec{x}}\,(k+1|k) = \hat{\vec{x}}\,(k|k) + \mathbf{L}(k)\left(\vec{y}(k) - \mathbf{C}\hat{\vec{x}}\,(k|k)\right) \tag{1.58}$$

$$\mathbf{P}\,(k+1|k) = \Big(\mathbf{I} - \mathbf{L}(k)\,\mathbf{C}\Big)\mathbf{P}\,(k|k) \tag{1.59}$$

where

$$\mathbf{A}(k|k-1) = \mathbf{I} + \tilde{\mathbf{A}}_1 t_s + \hat{\omega}(k|k-1)\tilde{\mathbf{A}}_2 t_s + \hat{\psi}_\alpha^r(k|k-1)\tilde{\mathbf{A}}_3 t_s + \hat{\psi}_\beta^r(k|k-1)\tilde{\mathbf{A}}_4 t_s \tag{1.60}$$

and where

$$\mathbf{B} = \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 & 0 & 0 \end{bmatrix}^T \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{1.61}$$

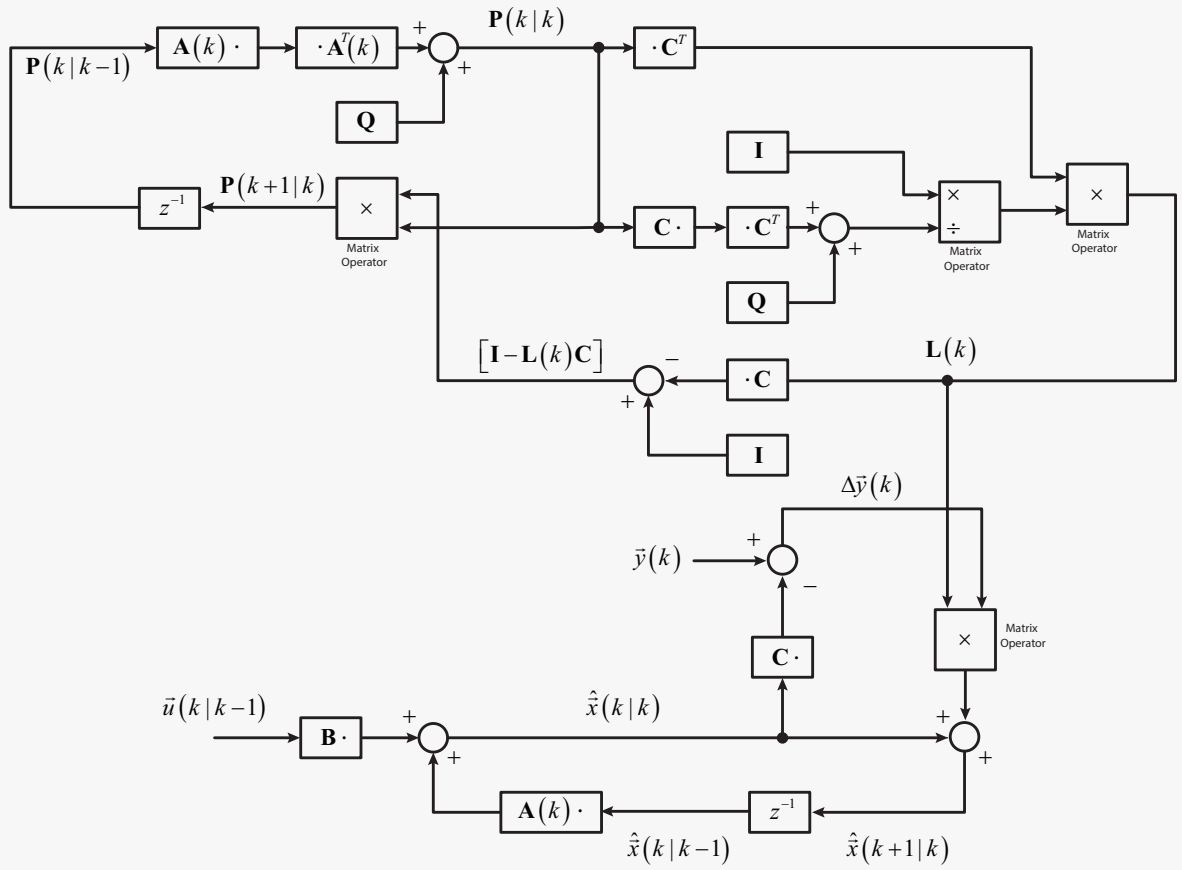


Figure 3: Kalman observer with time varying matrix transition.

1.2 EKF Observer for nonlinear PMSM

In additional to the previous EKF observer, we consider now the case where the torque expression is taken into account into the linearization along the observed state trajectory.

$$\frac{d}{dt}\vec{x}(t) = \vec{g}(\vec{x}) \Rightarrow \begin{cases} \frac{d}{dt}i_\alpha(t) = -\frac{R_s}{L_s}i_\alpha(t) + \frac{1}{L_s}\omega(t)\psi_\beta^r(t) + \frac{1}{L_s}u_\alpha(t) \\ \frac{d}{dt}i_\beta(t) = -\frac{R_s}{L_s}i_\beta(t) - \frac{1}{L_s}\omega(t)\psi_\alpha^r(t) + \frac{1}{L_s}u_\beta(t) \\ \frac{d}{dt}\psi_\alpha^r(t) = -\omega(t)\psi_\beta^r(t) \\ \frac{d}{dt}\psi_\beta^r(t) = \omega(t)\psi_\alpha^r(t) \\ \frac{d}{dt}\vartheta(t) = \omega(t) \\ \frac{d}{dt}\omega(t) = \frac{p}{J}\left\{\frac{3}{2}p\left[\psi_\alpha^r(t)i_\beta(t) - \psi_\beta^r(t)i_\alpha(t)\right] - \tau_l(t) - \frac{b}{p}\omega(t)\right\} \end{cases} \tag{1.62}$$

where we assumed $L_\alpha = Ł_\beta = L_s$ and $\frac{1}{p}\omega(t) = \omega_m(t)$.

$$\begin{aligned} \left.\frac{\partial}{\partial\vec{x}}\vec{g}(\vec{x})\right|_{\vec{x}=\hat{\vec{x}}} &= \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \cdots & \frac{\partial g_n}{\partial x_n} \end{bmatrix}_{\hat{\vec{x}}(t)} \\ &= \begin{bmatrix} -\frac{R_s}{L_s} & 0 & 0 & \frac{\hat{\omega}}{L_s} & 0 & \frac{\hat{\psi}_\beta}{L_s} \\ 0 & -\frac{R_s}{L_s} & -\frac{\hat{\omega}}{L_s} & 0 & 0 & -\frac{\hat{\psi}_\alpha}{L_s} \\ 0 & 0 & 0 & -\hat{\omega} & 0 & -\hat{\psi}_\beta \\ 0 & 0 & \hat{\omega} & 0 & 0 & \hat{\psi}_\alpha \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -\frac{3}{2}\frac{p^2}{J}\hat{\psi}_\beta & \frac{3}{2}\frac{p^2}{J}\hat{\psi}_\alpha & \frac{3}{2}\frac{p^2}{J}\hat{i}_\beta & -\frac{3}{2}\frac{p^2}{J}\hat{i}_\alpha & 0 & -\frac{b}{J} \end{bmatrix} \\ &= \tilde{\mathbf{A}}_1 + \hat{\omega}(t)\tilde{\mathbf{A}}_2 + \hat{\psi}_\alpha^r(t)\tilde{\mathbf{A}}_3 + \hat{\psi}_\beta^r(t)\tilde{\mathbf{A}}_4 + \hat{i}_\alpha(t)\tilde{\mathbf{A}}_5 + \hat{i}_\beta(t)\tilde{\mathbf{A}}_6 \end{aligned} \tag{1.63}$$

where $\hat{\vec{x}}$ is the estimated state vector from the EKF.



The matrices $\tilde{\mathbf{A}}_1$, $\tilde{\mathbf{A}}_2$, $\tilde{\mathbf{A}}_3$, $\tilde{\mathbf{A}}_4$, $\tilde{\mathbf{A}}_5$ and $\tilde{\mathbf{A}}_6$ are as follows

$$\tilde{\mathbf{A}}_1 = \begin{bmatrix} -\frac{R_s}{L_s} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{R_s}{L_s} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\frac{b}{J} \end{bmatrix}, \quad \tilde{\mathbf{A}}_2 = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{L_s} & 0 & 0 \\ 0 & 0 & -\frac{1}{L_s} & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{1.64}$$

$$\tilde{\mathbf{A}}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{L_s} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{2}\frac{p^2}{J} & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \tilde{\mathbf{A}}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{L_s} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{2}\frac{p^2}{J} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{1.65}$$

$$\tilde{\mathbf{A}}_5 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3}{2}\frac{p^2}{J} & 0 & 0 \end{bmatrix}, \quad \tilde{\mathbf{A}}_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{2}\frac{p^2}{J} & 0 & 0 & 0 \end{bmatrix} \tag{1.66}$$

The discretization of the transition matrix $\tilde{\mathbf{A}}(t)$ can be approximated by Taylor as follows

$$\begin{aligned} \mathbf{A}(k|k-1) = \mathbf{I}_6 + \tilde{\mathbf{A}}_1 t_s + \hat{\omega}(k|k-1)\tilde{\mathbf{A}}_2 t_s + \\ + \hat{\psi}^r_\alpha(k|k-1)\tilde{\mathbf{A}}_3 t_s + \hat{\psi}^r_\beta(k|k-1)\tilde{\mathbf{A}}_4 t_s + \\ + \hat{i}_\alpha(k|k-1)\tilde{\mathbf{A}}_5 t_s + \hat{i}_\beta(k|k-1)\tilde{\mathbf{A}}_6 t_s. \end{aligned} \tag{1.67}$$